PRIVACY-PRESERVING FEDERATED LEARNING FRAMEWORK FOR MULTI-HOSPITAL MEDICAL DATA ANALYSIS

CKD Case Study — Federated Learning with Hospital-wise Missing Features and Adaptive Privacy Protection.

Important: Hospitals in this prototype are simulated clients created from public datasets. The missingness-to-noise mapping is a proposed experimental mechanism and is not presented as a clinically established rule.

1. Datasets

• UCI Chronic Kidney Disease Dataset — ID 336
• Kaggle Chronic Kidney Failure Dataset Reboot
• UCI Risk Factor Prediction of Chronic Kidney Disease — ID 857

2. Simulated Hospital Configuration

Hospital	Dataset	Missing	Adaptive Noise
Hospital 1	UCI CKD 336	5%	0.05
Hospital 2	Kaggle Reboot — Part A	15%	0.10
Hospital 3	Kaggle Reboot — Part B	25%	0.20
Hospital 4	UCI Risk Factor 857	40%	0.30

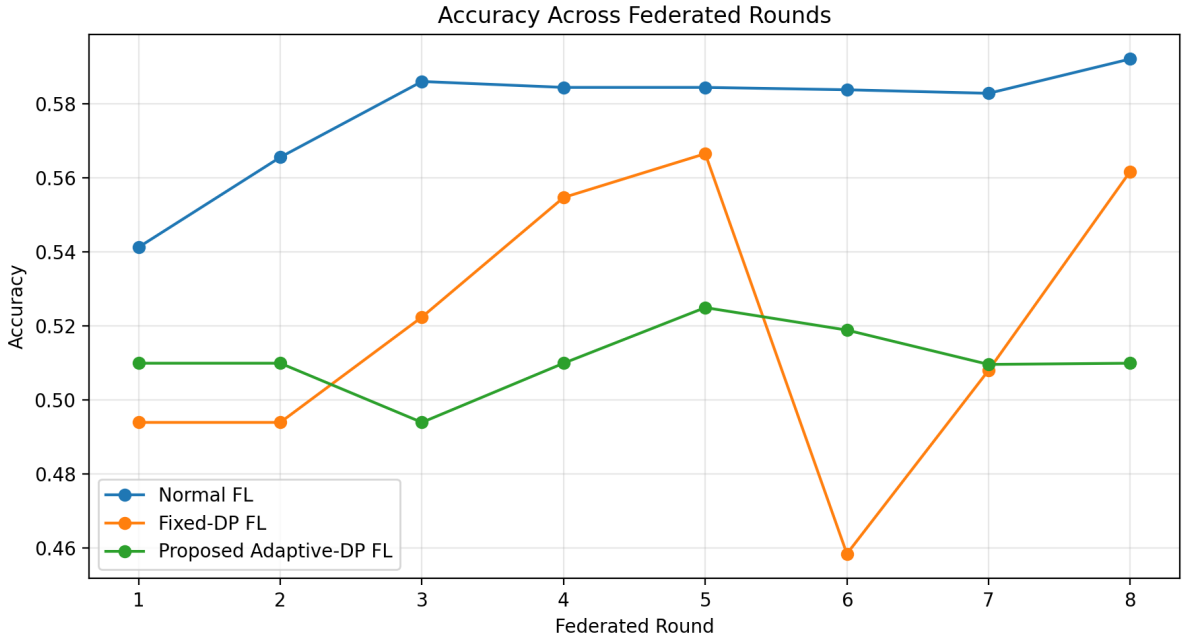
3. Final Model Performance

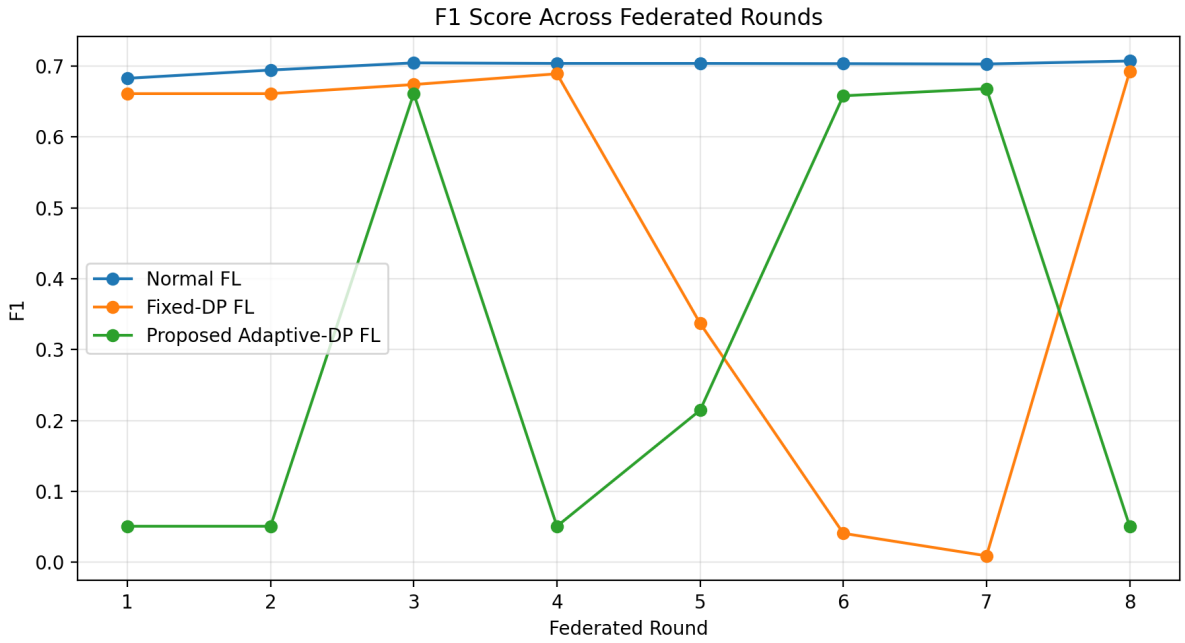
Method	Accuracy	Precision	Recall	F1	ROC-AUC
Normal FL	0.5920	0.5477	0.9981	0.7073	0.5928
Fixed-DP FL	0.5617	0.5299	0.9981	0.6922	0.5554
Proposed Adaptive-DP FL	0.5099	0.5857	0.0265	0.0507	0.5339

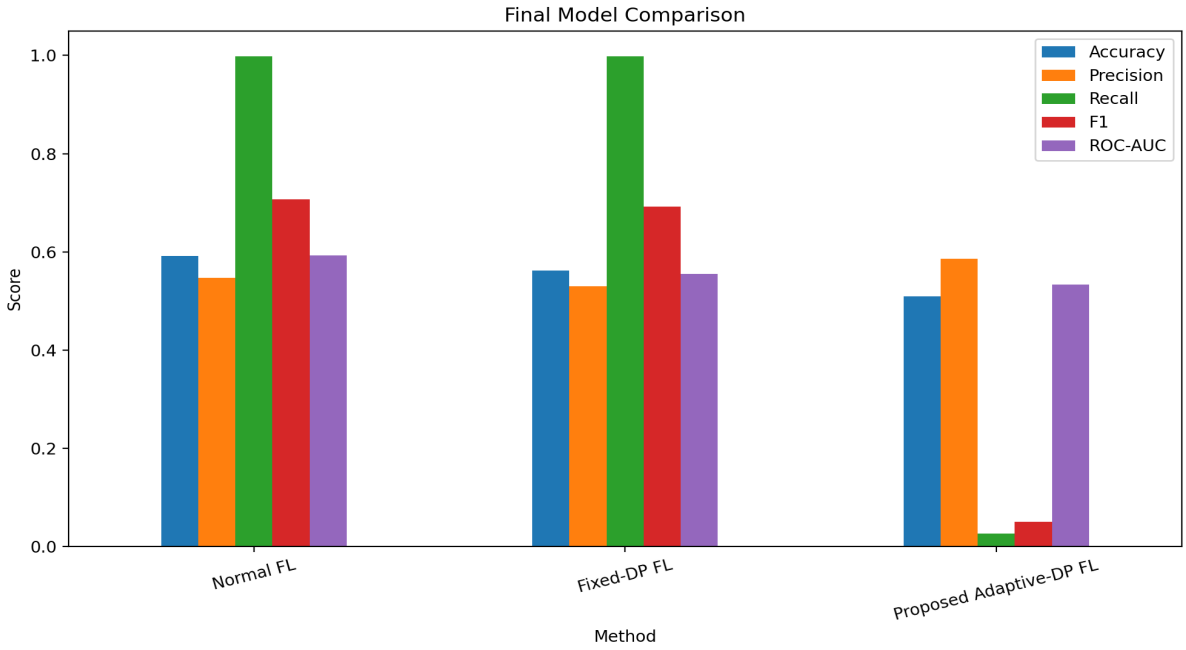
4. Privacy Evaluation — Confidence-based Membership Inference Attack

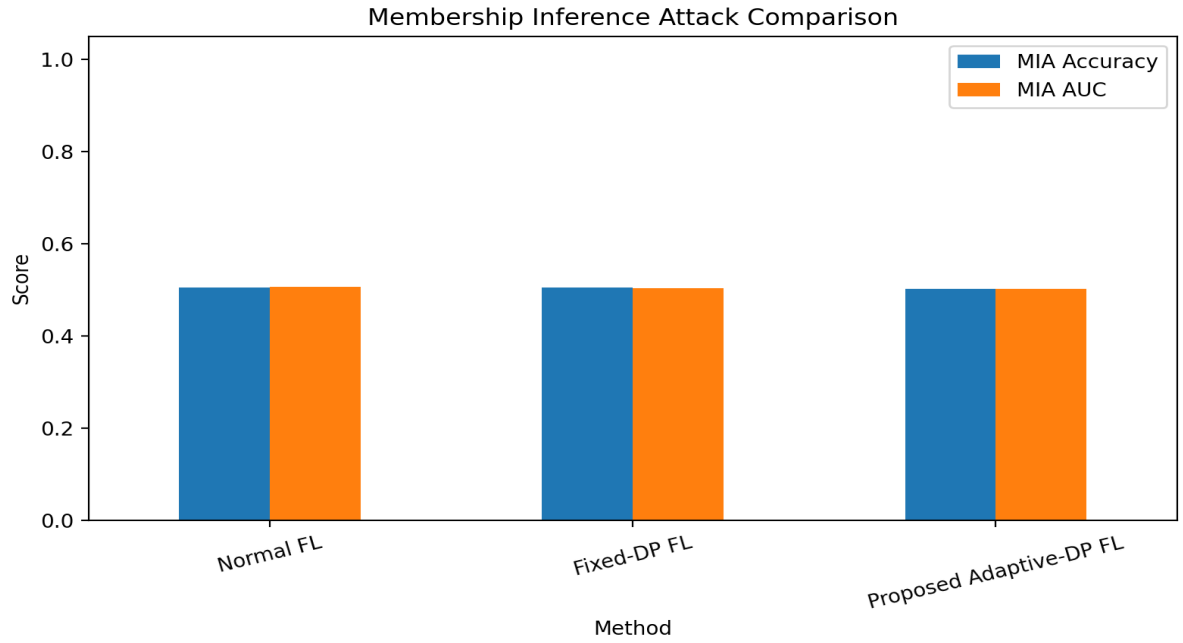
Method	MIA Accuracy	MIA AUC
Normal FL	0.5056	0.5066
Fixed-DP FL	0.5054	0.5037
Proposed Adaptive-DP FL	0.5021	0.5019

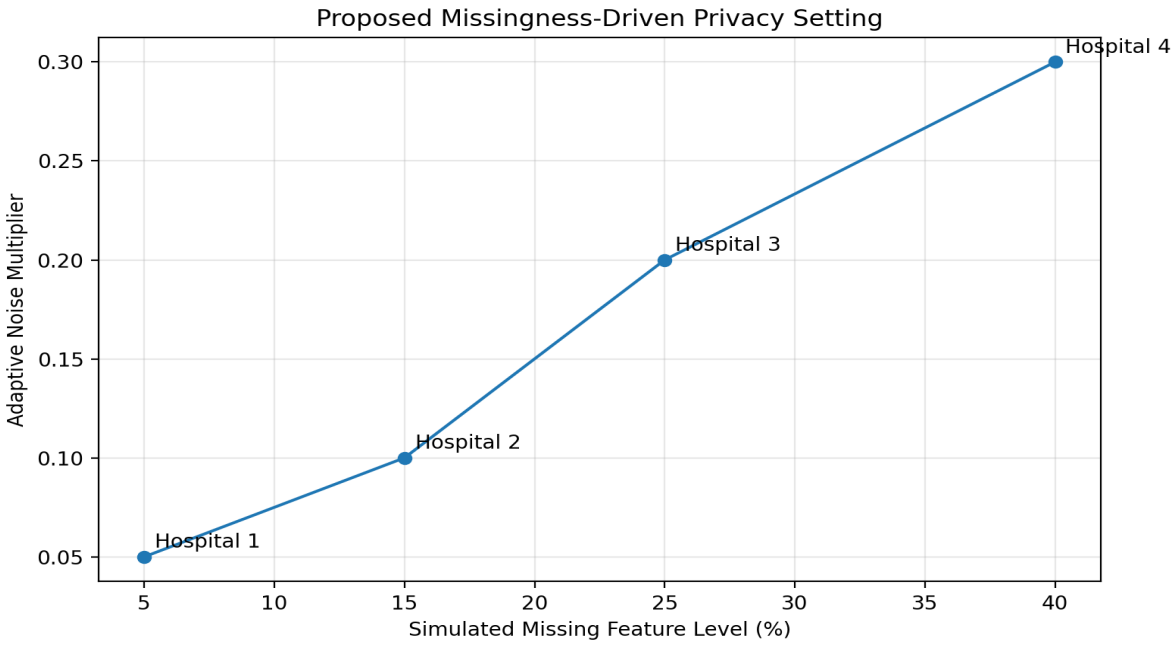
5. Graphical Results

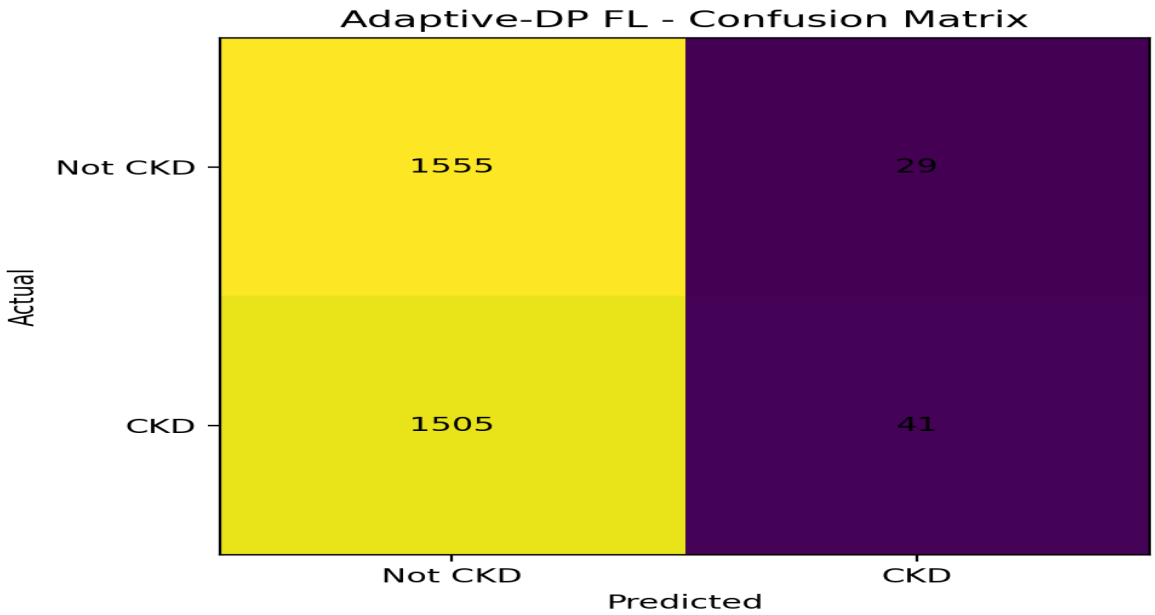

Accuracy Across Federated Rounds
Accuracy
Federated Round
Normal FL
Fixed-DP FL
Proposed Adaptive-DP FL
0.46
0.48
0.50
0.52
0.54
0.56
0.58
1
2
3
4
5
6
7
8


F1 Score Across Federated Rounds
F1
Federated Round
Normal FL
Fixed-DP FL
Proposed Adaptive-DP FL
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
1
2
3
4
5
6
7
8


Final Model Comparison
Accuracy
Precision
Recall
F1
ROC-AUC
Score
1.0
0.8
0.6
0.4
0.2
0.0
Normal FL
Fixed-DP FL
Proposed Adaptive-DP FL
Method


Membership Inference Attack Comparison
MIA Accuracy
MIA AUC
Score
1.0
0.8
0.6
0.4
0.2
0.0
Normal FL
Fixed-DP FL
Proposed Adaptive-DP FL
Method





6. Conclusion

This prototype demonstrates a privacy-preserving federated learning workflow for multi-hospital medical data analysis using CKD as the case study. Raw patient records remain within simulated hospital clients. Hospital-wise missing-feature levels are assessed and used by the proposed adaptive mechanism to select a privacy-noise setting. Protected local model updates are aggregated using Federated Averaging (FedAvg). Model performance is evaluated using Accuracy, Precision, Recall, F1-score and ROC-AUC, while privacy leakage is explored using a confidence-based membership inference attack prototype.